INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

DISPENSER, CARBONATED BEVERAGE, MECHANICALLY REFRIGERATED (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers mechanically refrigerated, pressurized, carbonated beverage (soft drink) dispensers for Naval shipboard use. The dispensers are for use in self-service serving lines to mix, cool, and dispense carbonated and nonpulp noncarbonated drinks. The carbonators in each unit are used in conjunction with CO_2 gas cylinders, which must be procured separately.

2. CLASSIFICATION

Size 1: Four dispensing heads
Size 2: Five dispensing heads

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The carbonated beverage dispenser shall be mechanically refrigerated, have a pressurized syrup feed system, and single-line serving. The dispenser shall be equipped with a key lock switch. The cabinet shall be provided with four 4-inch high stainless steel legs suitable for mounting the equipment to a countertop.

3.1.1 Refrigeration unit. The refrigeration unit shall consist of a compressor, an air-cooled type condenser, a pump, an ice bank or cold plate (when used), a carbonator, and electrical and plumbing connections. The refrigeration unit shall be completely self-contained, precharged, and wired with a hermetically sealed compressor unit. The unit shall be capable of maintaining a continuous drink temperature, at a rate of 600 6-ounce drinks in a 90-minute period, of not greater than 42°F without ice at the dispensing head when the ambient temperature is not greater than 90°F. The unit shall operate at water pressures from 15 to 80 pounds per square inch gauge.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson-Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 7310

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3.1.1.1 Refrigerant (ozone depleting substance). The refrigerant shall be in accordance with ASHRAE 15, Group A1 (nonflammable), be in accordance with ARI 700 (purity), and be approved by the Environmental Protection Agency under the Significant New Alternate Program. The refrigerants permitted for use, in order of preference, shall be of one of the following: R134A, R-401A, or R-401B.

3.1.2 Dispenser cabinet. The cabinet shall be of materials and components of the quality normally used by the manufacturer in its standard commercial dispensers, provided that the completed dispenser is in accordance with the provisions of this document. The dispensing assembly shall be constructed of stainless steel. Dispensing valves shall be electrically operated and shall be equipped with solenoids rated for continuous duty. Each valve shall be easily removable with basic hand tools without interrupting the operation of the other dispensing valves. Easily accessible connections with quick disconnect ball lock fittings shall be provided for cold water, syrup, and carbon dioxide supplies, and the fittings shall be noninterchangeable.

3.1.3 Hoses. Supply lines shall be the manufacturer’s standard commercial high-pressure reinforced hoses.

3.1.4 Drain pan. A removable drain pan and perforated tray shall be provided under the dispensing valves. The drain pan shall be removable without the use of tools and shall have a closeable drain hole.

3.1.5 Dispenser. The dispenser shall be a mechanically refrigerated, pressurized unit that dispenses carbonated beverages, noncarbonated beverages, and water. It shall dispense not less than 600 6-ounce drinks in a 90-minute period. The dispenser may be installed with syrup tanks, carbon dioxide tank, carbonator, and water supply, each remotely located not more than 50 feet from the dispenser. The dispensing heads shall be accessible from one side only. Two syrup tanks shall be provided for each dispenser head. Both tanks shall have the same capacity, either 3 or 5 gallons.

3.1.6 Physical size limitations. The cabinet shall pass or shall disassemble so that it can pass through a shipboard access measuring 26 x 66 inches with 8-inch radius corners. The refrigeration system shall not require disassembly to pass through access openings. When specified for submarines, the cabinet shall pass or shall disassemble so that it can pass through a 30 inch diameter submarine access. (Shipboard requirement, see 7.2)

3.2 Accessibility. The unit shall be physically and visually accessible from the front for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.3 Electrical requirements. The unit shall operate on 115 VAC, 60 Hz, single-phase power, as defined in MIL-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. Mercury-type float switches shall not be used. (Shipboard requirement, see 7.2)

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3.4 Inclined operation. The unit shall operate satisfactorily (in accordance with the requirements of this CID) with no spillage of liquid or product when test-operated for 30 seconds inclined at an angle of 15° (30° when specified for submarines) each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 7.2)

3.5 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167/1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 7.2)

3.6 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 18, UL 62, and UL 471.

3.7 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

3.7.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

3.7.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut-down.

3.8 Finish. The unit shall have an exterior finish of the manufacturer's standard commercial finish.

4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

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7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Size of dispenser required
- Capacity of syrup tanks, either 3 or 5 gallons
- When required, cold plate or ice bank
- Hose length required
- When required, manuals shall be in accordance with ASTM F 760

7.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

7.3 Source of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

MIL-STD-1399	-	Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating
MIL-STD-167/1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally excited)

7.3.2 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM F 760	-	Food Service Equipment Manuals

7.3.3 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 18	-	Manual Food and Beverage Dispensing Equipment

7.2.4 Underwriters Laborites (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 62	-	Flexible Cord and Fixture Wire
UL 471	-	Commercial Refrigerators and Freezers

7.2.5 American Refrigeration Institute (ARI) Standards. ARI Standards are available from the Air-Conditioning & Refrigeration Institute, 4301 North Fairfax Drive, Suite 425, Arlington, VA 22203.

ARI 700	-	Specifications for Fluorocarbons and Other Refrigerants

7.2.6 American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) Standards. ASHRAE Standards are available from the American Society of Heat, Refrigerating and Air-Conditioning Engineers, 1791 Tullie Circle, N.E., Atlanta, GA 30329-2305.

AHSRAE 15	-	Safety Code for Mechanical Refrigeration

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7.2.7 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

7.3 Suggested sources of supply. Manufacturers of products known to meet the requirements of this CID are listed below. However, competition is not limited to these companies.

ALCO Food Service Equip. Co.
455 E. Kehoe
Carol Stream, IL 60187

The Cornelius Co.
One Cornelius Place
Anoka, MN 55303-1592

MILITARY INTERESTS:

Military Coordinating Activity

Navy - SH

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity:
Navy - SH
(Project 7310-0893)

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